Physics-Informed LLM Latency Prediction:
Critical Analysis of the Analytical Roofline Model vs. Direct Operational Modeling

Pre-Execution Scheduling & The Token Count Prerequisite

In production LLM serving, latency predictors must operate **strictly pre-execution** using launch configurations: cluster parallelism (`tp`, `pp`, `dp`, `num_gpus`) and request budgets (`batch_size`, `output_tokens`). Post-execution telemetry is strictly excluded.

The Token Prerequisite (Imai et al., Section A.4): The analytical Roofline Model (RLM) calculates hardware ceilings from input and output tokens:

$$T_{\text{roofline}} = T_{\text{prefill}}(\text{input_tokens}) + \text{output_tokens} \times T_{\text{decode_step}}(\text{batch_size}) \tag{1}$$

While prompt length is known pre-execution, output length is dynamic. As acknowledged by Imai et al. (Appendix A.4): *"In practical application, the generation length is not predetermined... To accurately model inference latency in real world scenarios, we must consider predicting the generation length based on the input prompt."*

Methodological Analysis: Operational Modeling vs. Analytical Physics:

- **Data Aggregation**: To prevent duplicate data leakage across repeated benchmark trials, the 10 replicate runs per setting are aggregated into unique configuration means ($N = 24$ unique configurations for Qwen3-14B) prior to modeling.
- **Feature Collision (Why Base Fails)**: Without a workload indicator, identical launch configurations (`batch_size=128, out=500, DP2`) predict both Alpaca (≈ 54 ms) and Longbench (≈ 452 ms). This severe collision causes baseline R^2 to collapse ($R^2 < 0$).
- **Resolving Workloads via Categorical Task**: Providing a categorical task indicator (`task: Alpaca vs. Longbench`) resolves the collision, allowing tree models to achieve $R^2 \approx 0.63$.
- **Value of Analytical Roofline**: Following Imai et al., analytical Roofline latency is modeled strictly from hardware ceilings, batch size, and generation length without artificial prompt token assumptions. When combined with workload classification, it provides physical inductive bias that reduces mean squared error on tree ensembles by up to 18.2% (GB MSE drops from 109313 → 89363, RF MSE to 92138), while maintaining robust out-of-distribution generalization across unseen parallelism configurations (GroupKFold $R^2 \approx 0.51$).

Table 1: Qwen3-14B: Feature Set Ablation on Unique Configuration Means (R^2 / MSE).
Evaluated on 24 aggregated configuration cells via 3-Fold Cross-Validation. Base features: `[tp, pp, dp, num_gpus, batch_size, output_tokens]`.

Model / Estimator	Base (Collision)	Base + Categorical Task	Task + Analytical RLM	Key Takeaway
Linear Regression (LR)	-0.9465 / 373347	**0.2791 / 140298**	**0.2791 / 140298**	Task resolves collision
Random Forest (RF)	-0.7947 / 397206	0.6265 / 100881	**0.6413 / 92138**	RLM cuts MSE by 8.7%
Gradient Boosting (GB)	-1.5466 / 522026	0.5644 / 109313	**0.6100 / 89363**	RLM cuts MSE by 18.2%
Out-of-Distribution Generalization (+Roofline, GroupKFold across Unseen Parallelism)				
Linear Regression: $R^2 = 0.3708$ (MSE 135333); Random Forest: $R^2 = 0.5148$ (MSE 108194); Gradient Boosting: $R^2 = 0.4768$ (MSE 113426)				

Key Conclusions for Deployment & Next Steps

- **Unique Configuration Modeling**: Aggregating repeated physical trials ensures clean, leakage-free evaluation without artificial lookup-table memorization.
- **Workload Characterization**: In alignment with Imai et al., static token count assignments are removed. Explicit workload identifiers resolve feature collisions, and production deployments should capture continuous prompt distributions from telemetry.
- **Physics as an Inductive Prior**: Analytical Roofline latencies provide physical hardware scaling bounds that lower prediction variance and improve generalization across unseen cluster parallelisms ($R^2 \approx 0.51$).
- **Experimental Next Steps**: Scale evaluation across heterogeneous architectures (14B to 27B) on BSC MareNostrum 5 under continuous load distributions.